

4D-printed cementitious metamaterials with tunable vibration performance: Integrating programmable thermomechanics and forward prediction

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ABSTRACT

Low-frequency vibrations pose a significant challenge in urban environments due to their pervasive nature and limited effective control methods. This study presents a 4D-printed cementitious chain-bead acoustic metamaterial (CCAM) designed to mitigate low-frequency vibration in urban environments. The proposed CCAM leverages 4D printing to achieve a programmable thermomechanical response, enabling precise tuning of low-frequency bandgaps. Its tunability is achieved via two complementary methods including temperature sweep and geometric reconfiguration considering the dynamic thermal-mechanical nature of shape memory polymer. Bandgap behavior is characterized via both numerical and experimental methods. A multilayer perceptron predictor is also established to map between the design variables including temperature, geometric variables, and the established bandgap metrics. The developed forward model demonstrates high accuracy and robust generalization, serving as a computational tool for predicting the performance of tunable acoustic metamaterials under varying service conditions. The Pearson correlation analysis between each design variable and the bandgap metrics further elucidates their respective influence on the wave attenuation performance. This study may provide a data-driven intelligent design framework for smart cementitious metamaterials in urban vibration control, offering a robust and adaptable platform for developing advanced tunable acoustic systems in the construction sector.

1. Introduction

Low-frequency elastic waves serve risks to urban safety due to their diverse sources and strong destructive potential [1,2]. They may originate from a wide range of activities, including heavy machinery vibration [3], road traffic [4,5], construction work, and even natural events [6] such as earthquakes, all of which are intrinsically linked to urban environments. Moreover, these waves are enabled to long-distance propagation due to their long wavelengths and weak attenuation, where soils and buildings are readily penetrated with substantial energy retained [7]. Therefore, resonant amplification and cumulative damage are induced in infrastructures, manifested by cracking, accelerated

deterioration, and even collapses, with adverse effects on human comfort and health [8]. In response, mitigation of low-frequency vibration is widely pursued through passive and active strategies, including dampers [9–11], vibration-isolation materials [12,13], and active or semi-active control systems [14]. Despite demonstrated effectiveness, several drawbacks remain. In high-rise buildings, dampers have been demonstrated to effectively reduce vibration responses and enhance occupant comfort, but their added mass and spatial footprint increase extra load and space requirements [15], thereby complicating design under wind and seismic load cases. Similarly, effective suppression of low-frequency response is achieved at larger geometric scales for vibration-isolation materials [16,17], and the required increases in volume and stroke

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become critical determinants of feasibility. Moreover, conventional damping and isolation materials are tuned to specific, narrowband frequencies [18], and their performance may deteriorate sharply under off-design excitation. This inherently narrow frequency response significantly thus limits their applicability where vibrations often span a broad frequency spectrum. In summary, the current vibration suspension technologies invariably involve complex trade-offs among performance, size, tunable range, and design complexity [19]. Consequently, the development of lightweight, broadband and adaptively tunable mitigation strategy has emerged as a critical imperative in this field.

Acoustic metamaterials (AMs) may offer a promising approach to low-frequency vibration control [20–22]. Their distinctive periodic unit-cell arrangement compels incident waves to excite localized resonances [9,17], trapping vibrational energy, inhibiting propagation, and generating well-defined bandgaps at specific frequencies. This mechanism enables effective suppression of low-frequency elastic waves without excessive mass or volume [22,23], facilitating lightweight and compact designs even in the low-frequency regime [24]. An additional advantage lies in the high tunability of these bandgaps, affording remarkable design freedom [10,20,25]. By strategically altering geometric configuration, from ribbing [26] and labyrinthine pathways to key dimensional parameters [27,28], the bandgap distribution can be tailored on demand to target specific frequency ranges for suppression. For instance, Muhammad et al. [24] proposed a novel lattice with sinusoidal ligaments with local resonating modes, demonstrating that millimeter-scale unit cells can effectively suppress vibrations below 20 Hz. The close agreement between experiment and simulation also underscores an exquisite sensitivity of the bandgap distribution to geometric perturbations, whereby minimal geometric adjustments on AMs can dramatically reconfigure their bandgap distribution.

However, the practical application of AMs is ultimately limited by their fixed geometries [29,30]. Since most AMs are finalized during fabrication, their bandgap distributions are effectively fixed. Consequently, they are ill-suited for real-world environments where excitation frequencies are dynamic and unpredictable. To address this limitation, researchers pursue diverse methods to increase the tunability of bandgap distributions [31,32], including the use of new constituent materials [33,34] and data-driven design algorithms [35,36]. Dong et al. [37] introduced a new kind of cementitious metamaterials together with a co-design strategy that simultaneously adjusts material properties and architectural features to achieve markedly broader and more controllable bandgaps. Leveraging a data-driven design strategy, Gong et al. [35] developed a cementitious concentric star-shaped metamaterial (CCSM) for customizable vibration performance. This strategy facilitates both the forward prediction and inverse design of the metamaterial. Tao et al. [20] proposed a novel auxetic acoustic metamaterial and demonstrated that its bandgap variation range can be significantly enlarged by enhancing the auxetic behavior. Numerical results show that the structural bandgap can be effectively tuned through external loading-induced deformation, with a strain of 0.05 leading to an enhancement of the bandgap from 24.5% to approximately 28.5%. Nevertheless, the capabilities of these methods remain rooted in a static design paradigm, which proves inadequate for real-world dynamic conditions. The central challenge is the absence of truly intelligent metamaterial systems that can autonomously adapt their bandgap characteristics after deployment. Current designs are passive merely in their response mechanism, not in their operational intelligence. They cannot be pre-programmed with a behavioral repertoire for different stimuli. This critical limitation underscores the need for a new system engineered from the ground up with self-adaptive bandgap tunability [38].

The thermal-responsive nature of shape memory polymers (SMPs) [39,40] may be regarded as the key to establishing self-adaptive systems [41]. This unique behavior arises from their temperature-dependent mechanical properties [42–44], which change dramatically, allowing the metamaterial's configuration to be reconfigured and structurally

locked. The resultant geometric transformations [45,46], induced by temperature variations, consequently, lead to autonomous shifts in their bandgap characteristics [47,48], thereby significantly enhancing the tunability of the metamaterial's vibration attenuation performance. The unique thermomechanical behavior of SMPs, when combined with the design freedom of 3D printing, underpins the 4D printing technique. This advanced manufacturing approach enables printed structures to undergo pre-programmed shape or property transformations in response to targeted thermal stimuli. This active tuning mechanism may substantially expand the operational bandwidth of vibration control, opening avenues for intelligent metamaterial systems. Consequently, 4D printing enables the direct integration of both complex geometric programming and active, stimulus-responsive behavior into the metamaterial. However, the higher-dimensional design space may also lead to complex, non-unique mappings between target bandgap performance and required material programming and geometries, which poses a significant challenge for precise inverse design of SMP-based acoustic metamaterials. In this way, achieving precise control may also demand a more sophisticated design approach that is both predictive and computationally efficient.

This study introduces a 4D-printed cementitious chain-bead acoustic metamaterial (CCAM) for low-frequency vibration control. The approach leverages a 4D printing technique to achieve a programmable thermomechanical response, enabling precise tuning of low-frequency bandgaps. The bandgap tunability was expanded through two methods: temperature sweeps at a fixed configuration, and axial loading that induces geometric reconfiguration. A dynamic model and experimental measurements of transmission loss were used to characterize the bandgap behavior. The data pertaining to cross-temperature and geometric variations were employed to train a multilayer perception (MLP) model. This model maps the input parameters, including temperature T , rib thickness t , bead diameter d , and chain amplitude h , to bandgap metrics for forward prediction. The model's predictive accuracy and robustness were rigorously validated via five-fold cross-validation, while the correlations between input features and outputs were quantified using Pearson correlation analysis. A closed performance-driven design loop is thereby established, providing a robust basis for intelligent and adaptive bandgap control in cementitious acoustic metamaterials.

2. Materials and methods

2.1. Theory

When elastic waves are incident on the metamaterial, scattering and localized resonances are induced, leading to bandgap formation and inhibited propagation. This mechanism is illustrated in Fig. 1 by a simplified dynamic model. The system can be described by a generalized dynamic equilibrium equation, in which inertial and elastic restoring forces are balanced with the external excitation:

$$M\ddot{x} + Kx = f(t) \quad (1)$$

where M and K are the mass matrix and stiffness matrix of the periodic unit cell, respectively; x represents the time-dependent displacement vector and $\ddot{x}$ is its second-time derivative (acceleration); $f(t)$ denotes the external excitation vector in the time domain.

Assuming a harmonic solution $x(t) = Xe^{i\omega t}$ together with $f(t) = Fe^{i\omega t}$, the frequency-domain form becomes:

$$(K - \omega^2 M)X = F \quad (2)$$

where X and F are the complex amplitudes of displacement and external loading, respectively, and ω is the angular frequency.

When Bloch periodicity is imposed on the unit cell, the problem is reduced to a generalized eigenvalue form, as illustrated in Eq. (2).

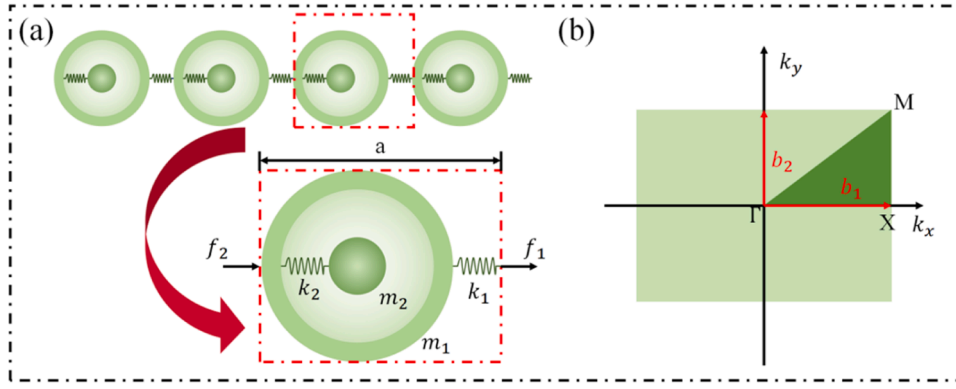


Fig. 1. Dynamic model of acoustic metamaterials: (a) local resonance model, (b) Brillouin zone.

$$K_r(q)U = w^2 M_r(q)U \quad (3)$$

or, equivalently,

$$\det[K_r(q) - w^2 M_r(q)] = 0 \quad (4)$$

Here, q denotes the Bloch wave vector in reciprocal space and U denotes the corresponding Bloch eigenvector, $K_r(q)$ and $M_r(q)$ are the Bloch-reduced stiffness and mass matrices obtained by enforcing Bloch boundary conditions on the unit cell.

Building on the generalized eigenvalue formulation in Eqs. (3)-(4), the periodicity of the structure is characterized by its real-space lattice vectors and their reciprocal counterparts. Specifically, the primitive lattice vectors are expressed as Eq. (5)

$$\begin{cases} e_1 = (\frac{p}{2}, 0) \\ e_2 = (0, \frac{p}{2}) \end{cases} \quad (5)$$

while the associated reciprocal vectors take the form as Eq. (6).

$$\begin{cases} b_1 = (\frac{\pi}{p}, 0) \\ b_2 = (0, \frac{\pi}{p}) \end{cases} \quad (6)$$

Here, p denotes the lattice pitch, namely the center-to-center spacing of neighboring unit cells. The e_1 and e_2 define the real-space primitive lattice, whereas b_1 and b_2 define its reciprocal counterpart, which sets the Brillouin zone. The IBZ, the minimal symmetry wedge in q -space, is

subsequently used for tracing dispersion relations.

2.2. Geometry

Fig. 2 depicts the geometric configuration of CCAM through the serial coupling of sinusoidal ligaments and circular beads. Within each unit cell, a central circular bead connects two mirror-symmetric sinusoidal chains, and each chain meets the cell corners through two quarter-circle beads. The CCAM unit cell was characterized by critical geometric parameters, including the rib thickness t , bead diameter d , and chain amplitude h . A metamaterial panel was subsequently generated by periodically tiling the unit cell in the plane, as a 4×4 array was adopted with four cells in both X and Y directions. For experimental implementation, two rectangular sheets with a thickness of 3 mm were attached to the upper and lower faces along the Y direction. Under this configuration, the overall CCAM geometry was defined. The in-plane dimensions were determined by the 4×4 unit-cell arrangement, yielding an overall height of approximately 244 mm and a width of 280 mm. Besides, an out-of-plane thickness of approximately 30 mm was obtained, and the thickness axis was oriented normal to the XY plane.

2.3. Materials and fabrication

The fabrication schematic of the CCAMs is depicted in Fig. 3. A bead-chain scaffold was fabricated from a commercial shape memory polymer filament via fused deposition modeling (FDM) employing an Ender-5 system, as illustrated in Fig. 3(a). The filament used has a nominal diameter of 1.75 mm, with a density of 1.19 g/cm³. During the printing

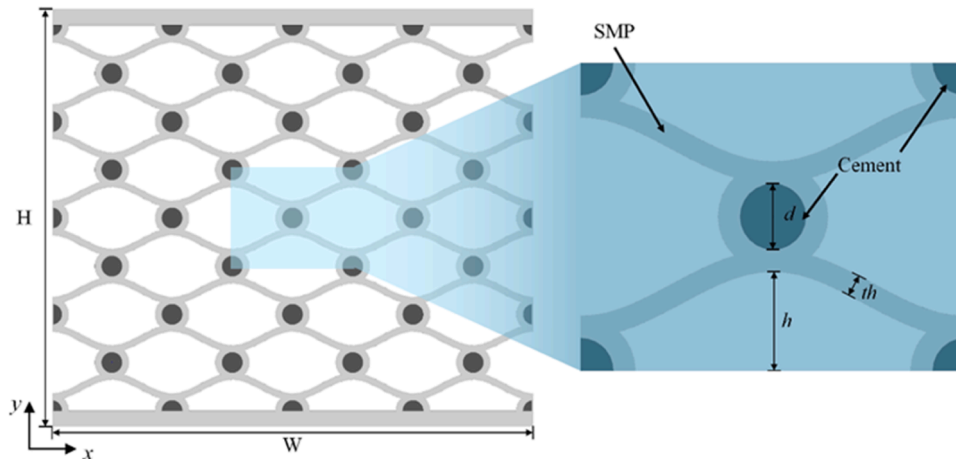


Fig. 2. Geometric schematic of the proposed CCAM.

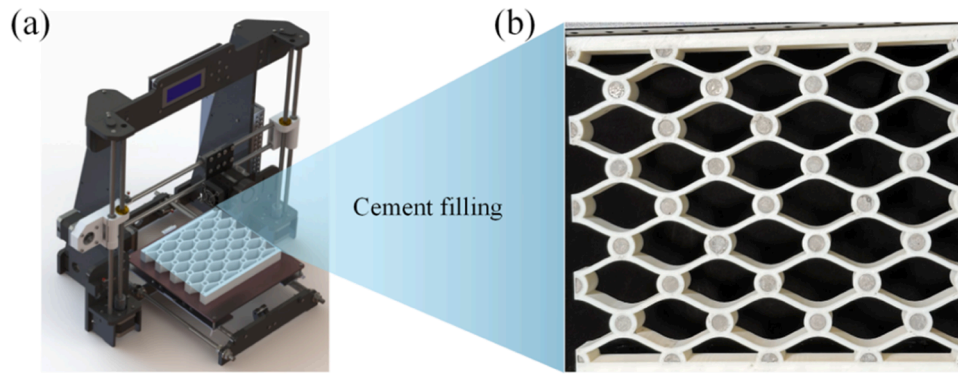


Fig. 3. Fabrication schematic of CCAM: (a) 3D printed SMP framework ; (b) the cementitious specimen.

process, the nozzle and bed temperatures were maintained at 220°C and 60°C, respectively. All specimens were produced with a layer thickness of 0.2 mm, a printing speed of 20 mm/s, and a nozzle diameter of 0.4 mm, achieving a full infill density of 100%. The thermo-mechanical properties of the shape memory filament were investigated by dynamic mechanical analysis (DMA+1000, Metravib, France). Test specimens were fabricated as depicted in Fig. 4, with dimensions of 30 mm × 5 mm × 0.9 mm. A temperature sweep test was conducted from room temperature (25°C) to 85°C under a sinusoidal strain with a frequency of 1 Hz and a strain amplitude of 5%. The thermal-mechanical properties of the SMPs, namely, the storage modulus, loss modulus, and loss factor ($\tan \delta$), are shown in Fig. 5(a) to be strongly temperature-dependent. The temperature-dependence may confirm that the SMP's stiffness can be actively regulated through thermal means. The glass transition temperature (T_g) is identified as 60.25°C from the loss factor peak, marking the critical transition between the glassy and rubbery states. Above this temperature, the SMP shall be transformed into a compliant, rubbery state, where a temporary shape can be imparted by an external force. This programmed shape is then effectively fixed into a stable, temporary configuration upon cooling below T_g . Similarly, the sharp decline of both modulus near T_g is attributed to the activation of polymer chain mobility. This transition provides the fundamental conditions for effective shape programming under external load. The stress-strain behaviour of SMP at different temperatures, such as 25°C, 55°C, 75°C, and 85°C, is also illustrated in Fig. 5(b). The figure shows that heating from 25°C to 85°C lowers the stress level and initial slope of the curves, indicating that the elevated temperature enhances chain mobility, leading to material softening and a transition from a glassy to a rubbery

state. Moreover, the SMP in this work exhibits a comparatively lower elastic modulus than those reported in previous studies. This is likely due to its lower crosslinking density and higher chain flexibility, both of which contribute to enhanced molecular mobility and reduced stiffness.

Following the fabrication of the SMP scaffold, the ordinary Portland cement (OPC) paste with a water-to-cement ratio of approximately 0.3 was prepared and cast into the scaffold, which also functioned as a mold. The cement, along with the integrated mold, was cured under standard laboratory conditions ($20 \pm 1^\circ\text{C}$, >95% RH) for 28 days prior to demolding. The cement part was subsequently characterized, and its density, as determined by the water displacement method, was measured to be around 1800 kg/m³. Finally, the resulting 4D-printed cementitious metamaterials, shown in Fig. 3(b), are composed of an SMP lattice framework infilled with cement paste.

2.4. Bandgap calculation

In this work, the theoretical analysis is based on a single CCAM unit cell under the ideal Bloch-Floquet boundary condition. The proposed theoretical and numerical models correspond to an infinite periodic array, making it suitable for capturing the intrinsic dispersion relations and bandgaps. Moreover, this study neglects interface debonding, manufacturing imperfections, and other secondary factors, focusing on its intrinsic performance metrics rather than constraints imposed by prototype dimensions and boundary conditions.

The dispersion characteristics of the proposed CCAM can be computed using eigenfrequency analysis in COMSOL Multiphysics. A two-dimensional plane-strain finite-element model of the CCAM unit

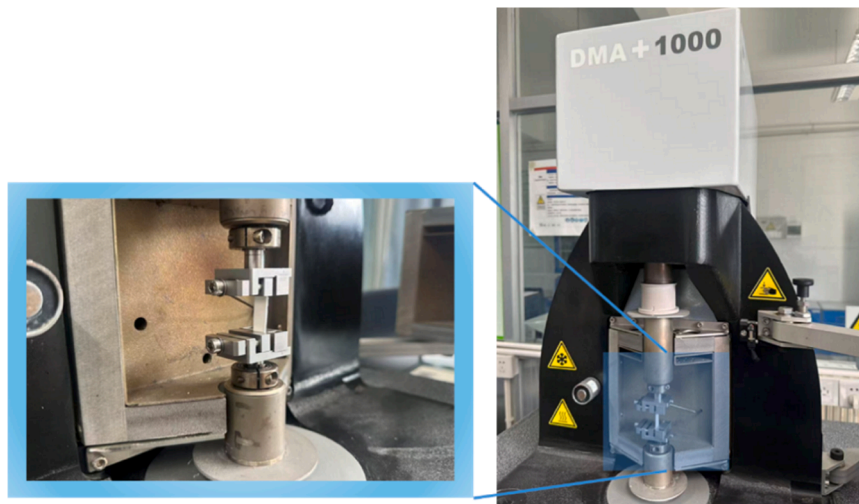


Fig. 4. The DMA setup and test specimens.

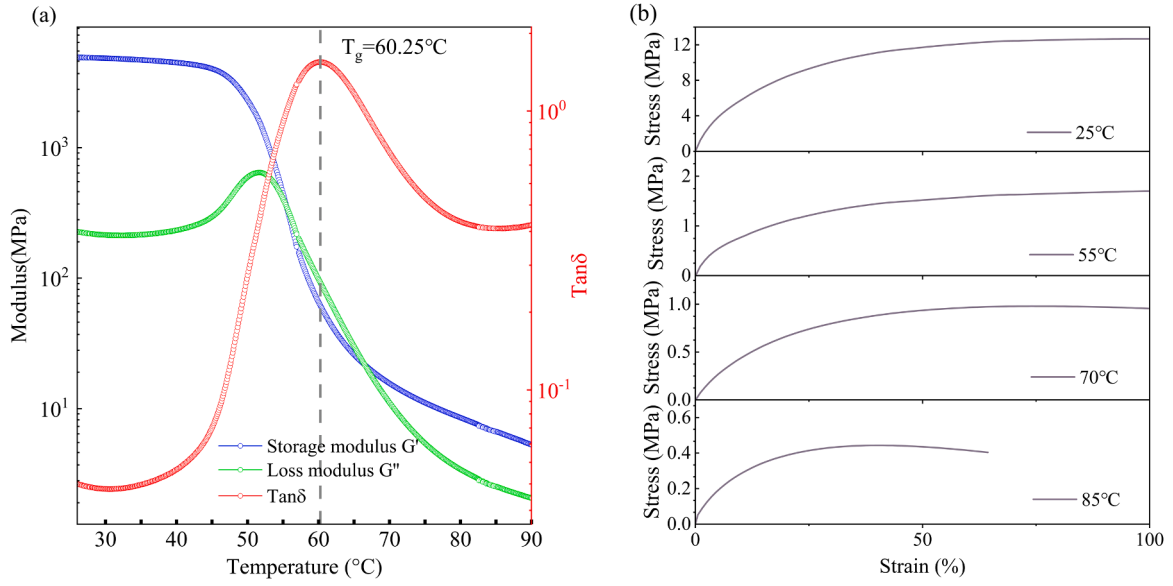


Fig. 5. Thermal-mechanical behaviour evaluation of SMP: (a) The storage modulus (G'), loss modulus (G''), and loss factor ($\tan \delta$) of SMP; and (b) stress-strain curves of SMP under different temperatures.

cell was constructed in the Solid Mechanics module. The geometry is discretized using a free triangular mesh as shown in Fig. S1, with convergence ensured through a systematic mesh refinement study until variations in the target eigenfrequencies fall below 1%. Bloch periodic boundary conditions were then imposed on the cell boundaries, and the Bloch wave vector q was swept along the edges of the IBZ to obtain the complete set of eigenfrequencies. The frequency-wave-vector relations were assembled into dispersion curves, from which bandgaps are identified as frequency intervals devoid of eigenmodes. To ensure the robustness of the results, the IBZ path (e.g., Γ -X-M- Γ) and a 0.01 of step size were finally determined where the locations and widths of the bandgaps remain stable within the convergence tolerance.

The bandgap performance of CCAMs can be subsequently evaluated by four critical metrics, including frequency ratio (η), normalized bandgap width ($\Delta\omega_{\text{nom}}$), average bandgap width ($\Delta\omega_{\text{avg}}$), and bandgap aggregation index (BGI). The η demonstrates the total fraction of the 0–3000 Hz frequency range that is occupied by all the bandgaps. It reads as follows :

$$\eta = \frac{\sum \Delta\omega}{3000} \quad (7)$$

where $\Delta\omega$ denotes width of each bandgap within 3000 Hz. The $\Delta\omega_{\text{avg}}$ represents the average width of all the m bandgaps within 3000 Hz, and it reads as :

$$\Delta\omega_{\text{avg}} = \frac{\sum \Delta\omega}{m} \quad (8)$$

The $\Delta\omega_{\text{nom}}$ denotes the normalized width of the largest (principal) bandgap, which quantifies how wide the principal bandgap is relative to its center frequency. And it can be expressed as follows :

$$\Delta\omega_{\text{nom}} = \frac{\omega_{\text{max}}^u - \omega_{\text{max}}^l}{\frac{\omega_{\text{max}}^u + \omega_{\text{max}}^l}{2}} \quad (9)$$

The BGI qualifies the joint extent and spectral compactness of bandgaps relative to the principal gap, and a larger BGI indicates that the gaps are both wider and more tightly clustered around the principal gap. Its expression is:

$$\text{BGI} = \sum \frac{\Delta\omega_i}{\Delta\omega_{\text{max}}} \left(1 - \frac{|\omega_{\text{max}}^{\text{cen}} - \omega_i^{\text{cen}}|}{\omega_{\text{max}}^{\text{cen}}} \right) \quad (10)$$

2.5. Transmission loss

The vibration suspension effect of the proposed CCAM can be measured through transmission loss evaluation. Different from theoretical analysis, the CCAM is fabricated as a finite panel configured as a 4×4 cell array, primarily due to constraints in manufacturing processes, experimental costs, and testing conditions. As illustrated in Fig. 6, the CCAM specimen was mounted inside a temperature chamber. After the chamber reached and stabilized at the setpoint ($\pm 0.1^\circ\text{C}$), a sine-sweep excitation (0–700 Hz) was generated by a pulse generator (Tektronix AFG3052C), amplified, and fed to a shaker (NUAA HEV-500). Inlet and outlet responses were measured with two accelerometers (Chengtec CT1010L), conditioned by a current amplifier (Chengtec CT5204), and recorded on an oscilloscope (Tektronix TDS2024C). To ensure consistent boundary conditions, the specimen's self-weight was neutralized by suspending it with fine and soft threads prior to attachment, thereby isolating the dynamic response from static gravitational effects. Finally, the transmission-loss performance of the CCAM can be qualified via the displacement-based frequency response function (FRF). The FRF is defined as:

$$\text{FRF} = 20 \log_{10} \frac{u_{\text{inl}}}{u_{\text{out}}} \quad (11)$$

where the u_{inl} and u_{out} are the excitation and the output displacements, respectively.

A numerical model was also established to verify the aforesaid experimental results. In this framework, wave propagation was simulated in a strip domain comprising a 1×10 array of CCAM unit cells (Fig. 7). To suppress spurious boundary reflections, perfectly matched layers (PMLs) were attached to the inlet and outlet. The PML employed a quadratic stretching profile and was parameterized to match the host medium's impedance at the physical-PML interface, thereby minimizing back-reflection. A unit-amplitude harmonic displacement was applied uniformly at the inlet surface, while the outlet responses were recorded at probe points. As illustrated in Fig. 7, the continuity (periodic) boundary conditions were applied between adjacent unit cells along the transverse direction to emulate an infinitely periodic domain. The numerical transmission-loss results can likewise be characterized by the FRF in Eq. (11), keeping consistency with the experimental measurements.

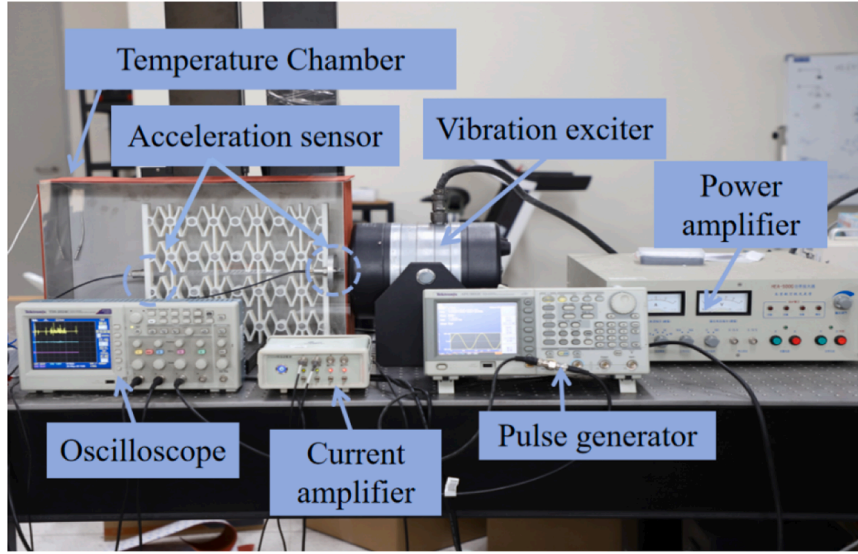


Fig. 6. Experimental setup on transmission loss.

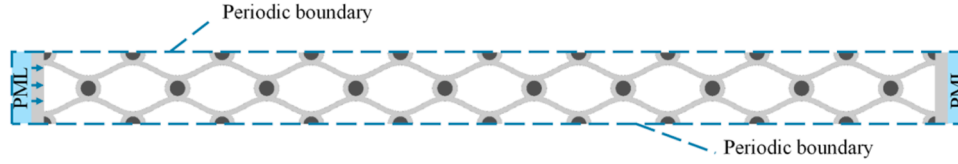


Fig. 7. Numerical model of transmission loss.

2.6. Thermomechanical-induced programmability

The CCAM geometry was programmed by thermomechanical loading to enable bandgap tuning. The relationship between deformation and bandgap response was established and quantified through coordinated numerical simulations and experiments. Experimentally, a 4×4 unit cell array specimen was fabricated and tested in a thermal loading system (WDW-50H, Wenteng, China). The protocol used speed loading at 1 mm/s under an isothermal environment of 75 °C to reach prescribed engineering strains from 0 to 0.4. While the strain was maintained, the specimen was cooled to 25 °C to fix the reconfigured architecture. The load was then released, and the transmission characteristics of the shape fixed CCAM were measured using the vibration test setup shown in Fig. 6.

Accordingly, the numerical analyses were conducted consistently in the above experimental conditions. The CCAM unit cell was discretized with quadratic tetrahedral elements and subjected to displacement loading, with temperature dependent properties for the SMP and the cement specified at 75 °C. The Quasi static loading then produced engineering strains of 0.1, 0.2, 0.3, and 0.4. At each strain level, the deformed configuration was reconstructed and evaluated in COMSOL Multiphysics by coupling the Solid Mechanics and Elastic Waves modules in the frequency domain. For the dispersion calculation, the shape fixed state was represented with 25 °C properties, and Bloch Floquet periodicity was enforced to obtain bandgap distribution through a linearized prestressed eigenfrequency analysis. In this way, experimental and numerical campaigns were designed to mutually verify that the bandgap can be programmed and tuned by thermomechanical loading.

2.7. Forward prediction

A regulation strategy for bandgap characteristics was formulated

that jointly considers temperature and geometry according to the dynamic thermomechanical response of the CCAM. As illustrated in Fig. 8, the MLP neural predictor was constructed to map the design parameters (t, d, h, T) to the target bandgap metrics, namely, η , $\Delta\omega_{avg}$, $\Delta\omega_{nom}$, and BGI . Specifically, the geometric design space was uniformly sampled to ensure broad coverage, with the temperature restricted to the four levels in Fig. 5. This specific set of levels was selected because the SMP density and elastic modulus had been experimentally characterized for them. The inputs were constrained in the range as illustrated in Eq. (12). And the prediction outputs follow Section 2.4 and comprise η , $\Delta\omega_{avg}$, $\Delta\omega_{nom}$, and BGI .

$$\left\{ \begin{array}{l} h \in (1 \text{ mm}, 6 \text{ mm}) \\ d \in (5 \text{ mm}, 8 \text{ mm}) \\ t \in (1 \text{ mm}, 6 \text{ mm}) \\ T \in (25^\circ\text{C}, 55^\circ\text{C}, 70^\circ\text{C}, 85^\circ\text{C}) \end{array} \right. \quad (12)$$

From each sample defined by h, t, d , and T , the temperature-dependent material properties were assigned. A periodic unit cell was then constructed in COMSOL and meshed, and Bloch–Floquet boundary conditions were imposed. The eigenvalue problem was solved along the irreducible Brillouin zone to obtain the dispersion curves. The bandgaps within the target frequency range were subsequently identified, their lower and upper edges were located, and mid-gap frequencies were computed. Finally, the bandgap metrics described in Section 2.4 were all evaluated for each sample.

A total of 2250 samples were subsequently generated. Of these, 250 were held out as an independent test set for the final generalization assessment, and the remaining 2000 were used for training. A hyperparameter search was then conducted on the 2000 training samples using 5-fold cross-validation. In each fold, roughly four-fifths of the samples (~ 1600 samples) were used for fitting and the remainder (~ 400

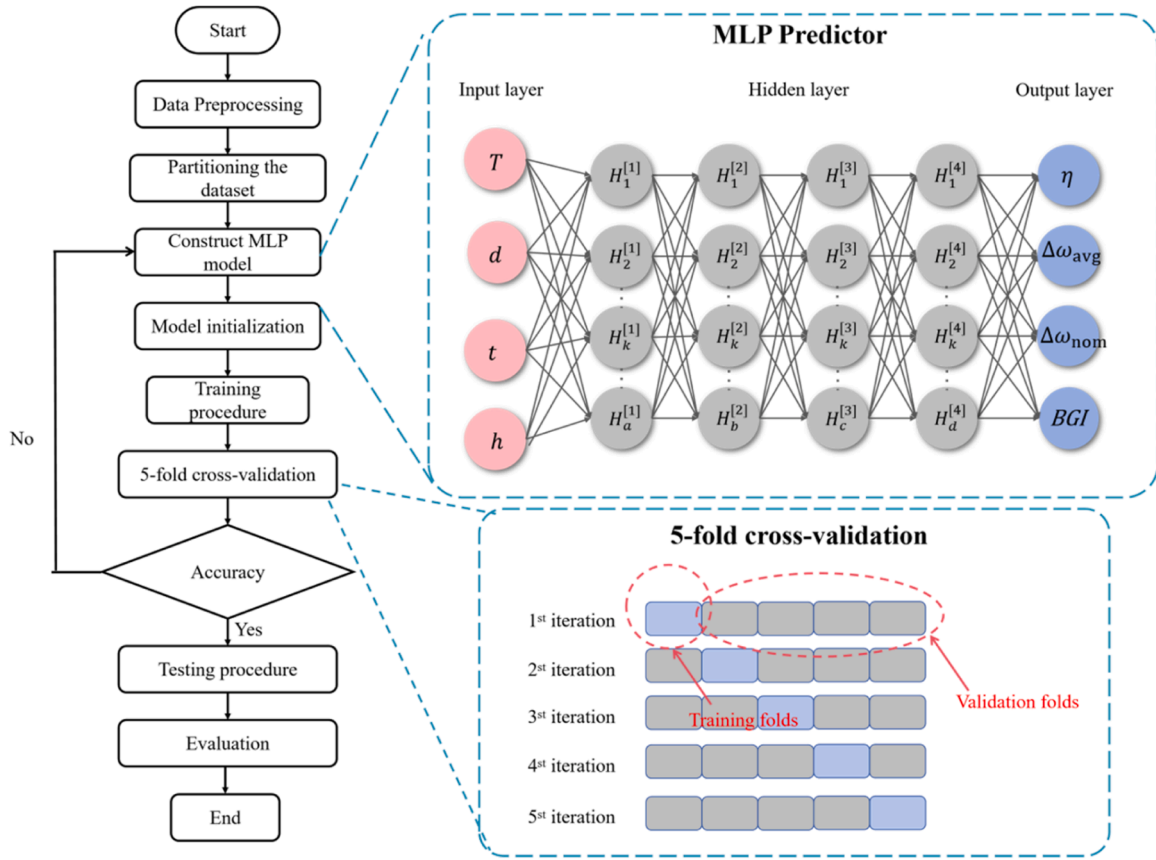


Fig. 8. The proposed framework for forward prediction model.

samples) for validation to tune hyperparameters and enable early stopping. Across the five folds, each sample served once as validation, yielding out-of-fold predictions. The prediction accuracy was summarized by the mean and the standard deviation across folds. Based on these results, the hyperparameters and a fixed training horizon were chosen. The network was then retrained on all 2000 training samples and evaluated once on the 250-sample test set. Under the selected configuration, a fully connected architecture was finally employed, comprising an input layer, four hidden layers with 22, 26, 55, and 32 neurons, and a linear output layer for regression. Hidden layers used the rectified linear unit (ReLU) activation defined as $\max(0, x)$, which introduces nonlinearity and helps mitigate vanishing gradients. Training minimized the mean-squared error with Adam at an initial learning rate of 1×10^{-3} , a batch size of 64, and a maximum of 300 epochs. Early stopping was governed by the validation Root Mean Square Error (RMSE) with a patience of 20. Light regularization was applied through dropout of 0.1 and L2 weight decay of 1×10^{-4} .

In this study, the accuracy of the proposed forward prediction model was evaluated using a complementary set of metrics, including RMSE, Mean Absolute Percentage Error (MAPE), and R^2 . Their expressions are as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - y_i)^2} \quad (13)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\tilde{y}_i - y_i}{y_i} \right| \quad (14)$$

where the $\tilde{y}_i$ and y_i are the prediction and numerical value of the i^{th} sample, respectively. And n is the number of samples in the datasets.

2.8. Correlation

To mitigate the black-box nature of the forward predictive model, Pearson correlation analysis was employed to quantify the linear dependencies between CCAM design variables and bandgap metrics. This approach may provide crucial insights into parametric interactions, improving the transparency of the design framework. The correlation coefficient r for the design variables X and bandgap metrics Y is defined by:

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \quad (15)$$

where σ_X and σ_Y denote the sample standard deviations, and $\text{cov}(X, Y)$ denotes the sample covariance.

3. Results and discussion

3.1. Vibration mode

Fig. 9 demonstrates the vibration mode of CCAM corresponding to its eigenfrequencies of 55.73 Hz and 184.53 Hz at 85 °C. At a frequency of 55.73 Hz, a localized torsional mode is distinctly excited within the structure. As shown in Fig. 9(a), the two upper-right quarter cement cores exhibit marked counterclockwise acceleration, which contrasts with the clockwise acceleration observed in the adjacent SMP ribs. A well-defined nodal line is consequently established along the interface between these components. The observed localized twisting is fundamentally attributed to a significant mismatch in stiffness and density between the cement and the SMP. Such material disparity resulted in incompatible deformation at their shared interface during dynamic excitation, thereby generating intense localized shear stresses that ultimately induced the antagonistic rotational motions.

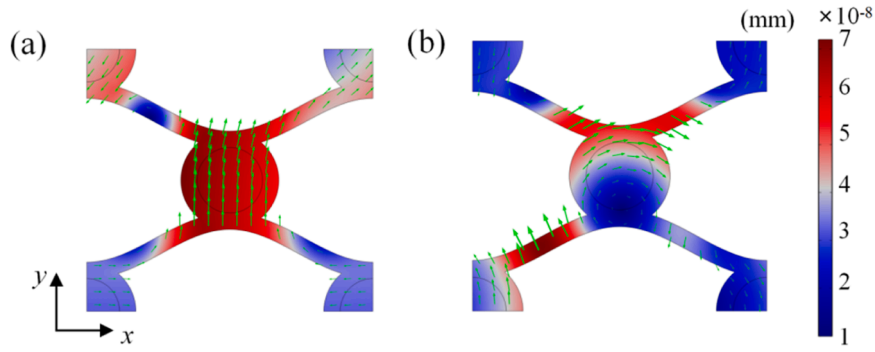


Fig. 9. Vibration modes of the CCAM under 85 °C at different frequencies: (a) 55.73 Hz and (b) 184.53 Hz.

The more complex mode shape is exhibited at the higher eigenfrequency of 184.53 Hz, featuring multiple regions and anti-direction motions among various components. Specifically, opposing accelerations were observed in the bottom quarter cement core regions. Concurrently, a nearly symmetric acceleration distribution is displayed by the SMP frame, indicating a coupling between global bending and local vibrations. Furthermore, reversed shear accelerations, resulting from abrupt changes in local stiffness, are identified at several rib-cement core connections. Collectively, these phenomena signify that vibrational energy is distributed more diffusely and modal complexity increases markedly at higher frequencies.

In summary, these vibration modes confirm the existence of material- and structure-governed localized resonance in the CCAM, paving the way for designing low-frequency vibration suppression strategies.

3.2. Bandgap distribution

Fig. 10 presents the distributions of the first four bandgaps in the CCAM at temperatures of 25°C, 55°C, 70°C, and 85°C. In Fig. 10(a), the second bandgap is identified as the principal one at 25°C. Its lower and upper boundaries are measured at 2549.17 Hz and 4434.90 Hz, respectively, confirming the effective suppression of mechanical wave propagation across this frequency range. In contrast, the first bandgap exhibits a much narrower bandwidth of only 19 Hz, located between 1254.13 Hz and 1273.13 Hz at the same temperature of 25°C. The third and fourth bandgaps show intermediate characteristics, with bandwidths of approximately 608.79 Hz and 803.85 Hz, respectively.

A consistent phenomenon is also observed when comparing bandgap distributions at elevated temperatures with the reference 25°C case. As

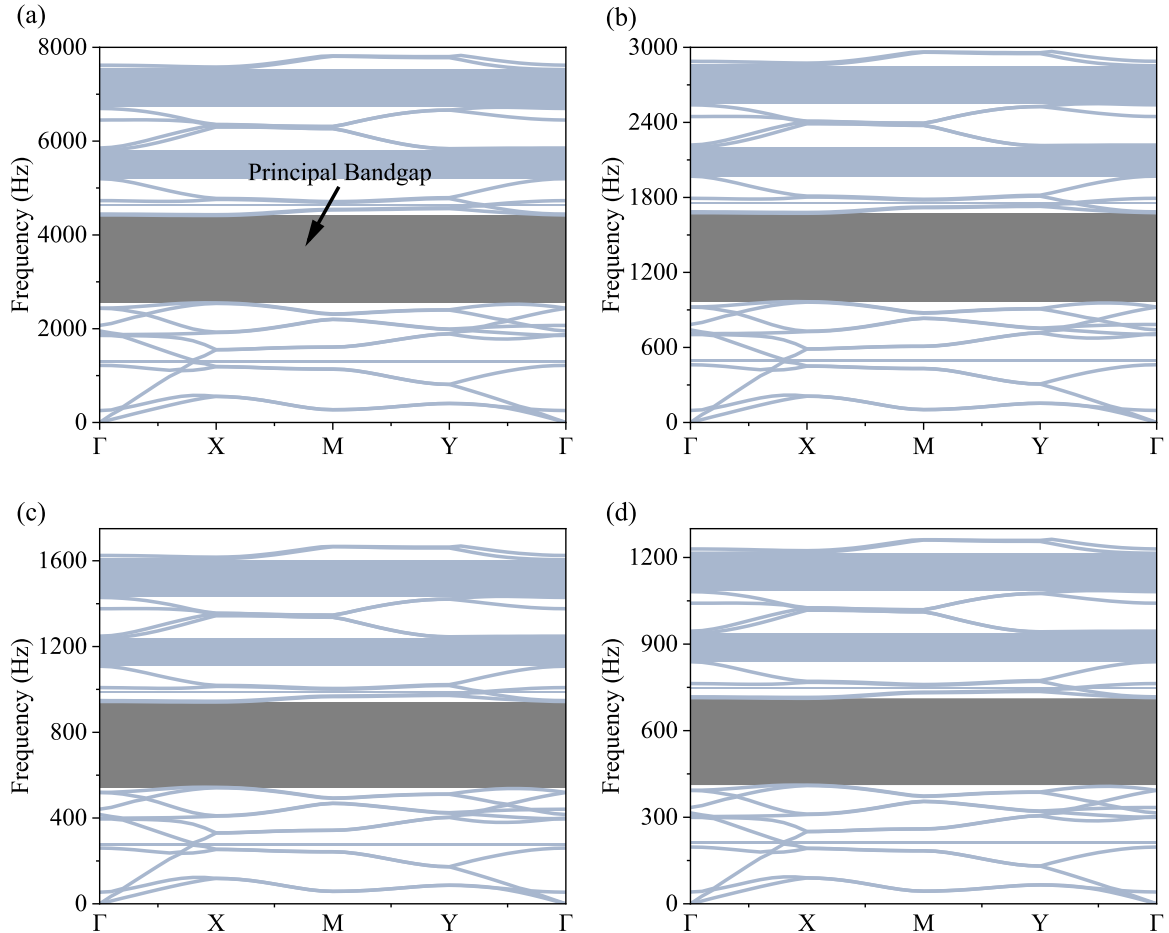


Fig. 10. Bandgap distribution of CCAM at various temperatures: (a) 25°C, (b) 55°C, (c) 75°C, and (d) 85°C.

the temperature rises, the upper and lower boundary frequencies of each bandgap shift toward lower frequencies, while their absolute bandwidths remain unchanged. For instance, when the temperature is increased from 55°C (Fig. 10(b)) to 85°C (Fig. 10(d)), the frequency range of the principal bandgap is reduced from 966.84 to 1681.64 Hz to 411.61–715.21 Hz, while its bandwidth is maintained consistently at approximately 715 Hz. The observed frequency shift arises from a uniform temperature-dependent scaling of the CCAM's eigenfrequencies. In contrast, bandgap width is governed by different mechanisms, primarily modal coupling, which remains insensitive to thermal changes. The underlying principle is that while temperature alters the phase accumulation in wave propagation, it does not affect the relative impedance contrast responsible for the bandgap formation. Consequently, the bandgap width remains stable, demonstrating that temperature can reliably tune the bandgap position without compromising its integrity, thus providing effective suppression of low-frequency elastic waves. It ultimately provides a strategic approach for tuning low-frequency vibration suppression in the proposed CCAMs.

3.3. Frequency response

Fig. 11 presents the transmission loss of the CCAM at 85°C covering a range of frequencies. Markedly high transmission loss is observed at 600 Hz and 900 Hz, where elastic waves are strongly attenuated and cannot propagate effectively. This behavior confirms that such frequencies lie within the CCAM's full bandgap, which shows excellent agreement with the bandgap distributions in Fig. 10(d). Conversely, the CCAM exhibits low transmission loss at frequencies such as 250 Hz, 300 Hz, and 450 Hz, permitting elastic waves to propagate with minimal attenuation. This result is consistent with the passband regions and collectively confirms the frequency-selective suppression capability of the structure. The correlation between the transmission spectrum and the bandgap distribution collectively establishes a coherent physical picture of the CCAM's elastic wave manipulation capabilities. Moreover, the physical mechanism underlying this frequency-selective suppression has been widely corroborated by related studies. Although specific structural designs may differ, previous work consistently confirms that targeted suppression of elastic waves in certain frequency ranges can be reliably achieved through rational structural design, underscoring the universality and practicality of this approach.

3.4. Geometric programmability

Fig. 12 illustrates the coordinated evolution of vibration modes in the CCAM under thermal-mechanical coupling as strain varies. At a

strain level of 0.1, shown in Fig. 12(a), the principal bandgap forms between 449 Hz and 761.8 Hz. The vibration response features local twisting patterns resulting from clear decoherence in acceleration directions, as indicated by the red arrows in Fig. 13(a). Such a specific dynamic response may serve as direct evidence of the local resonance mechanisms governing bandgap generation at these spectral positions. When the strain is increased to 0.4, as shown in Fig. 12(d), the bandgap range shifts to 562.03–702.89 Hz. This leads to a complete reorganization of the vibration mode into a new spatial pattern (Fig. 13(d)), with both acceleration directions and phase relationships now arranged in configurations distinctly different from the initial state. Such correlated evolution demonstrates that strain modulates the bandgap by dynamically restructuring the system's response.

Fig. 13 reveals the strain-dependent bandgap characteristics of CCAM. At a strain of 0.1, five complete bandgaps emerge within the 0–1600 Hz range. The principal bandgap already attains a width of 312.8 Hz, demonstrating that bandgap reorganization initiates under minimal strain conditions. With increasing strain, continuous nonlinear evolution is observed. As the strain increases from 0.2 to 0.3, the number of bandgaps further consolidates from four to three. During this process, the central frequency of the principal bandgap continues to shift upward while its width first expands to 340 Hz and then to 374 Hz, accompanied by a reduction in the bandgap ratio. The evolution culminates at 0.4 strain, where the bandgap distribution undergoes significant reorganization. Although four bandgaps are restored, all exhibits are reduced widths, including a decrease in the principal bandgap to 140.85 Hz.

Overall, the vibration suspension performance of CCAM is directly controlled by applied strain through geometric reconfiguration, enabling precise adjustment of both the vibration mode and bandgap distribution within the 0–1600 Hz range for programmable vibration control.

3.5. Prediction results

Fig. 14 denotes the predictive performance and error distribution for four key bandgap metrics of the CCAM. A high degree of statistical consistency is observed between the predicted and numerical values, as evidenced by R^2 values exceeding 97.5% across all evaluated metrics, confirming its high prediction accuracy and robust generalization. Although a maximum absolute error of -0.032 is observed at sample 144 in the prediction of bandgap ratio η (Fig. 14(b)), this deviation represents merely 3.2% of the total spectral range. The model is further demonstrated by its exceptional overall prediction accuracy ($R^2 \sim 0.99$) on the test set. The consistent performance across diverse samples, with only minimal errors in isolated cases, confirms the model's robust

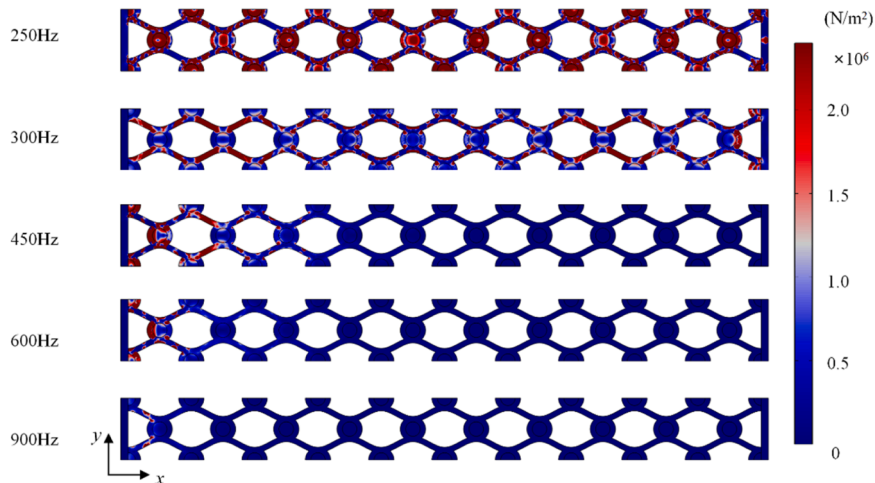


Fig. 11. Frequency response of CCAM at different frequencies.

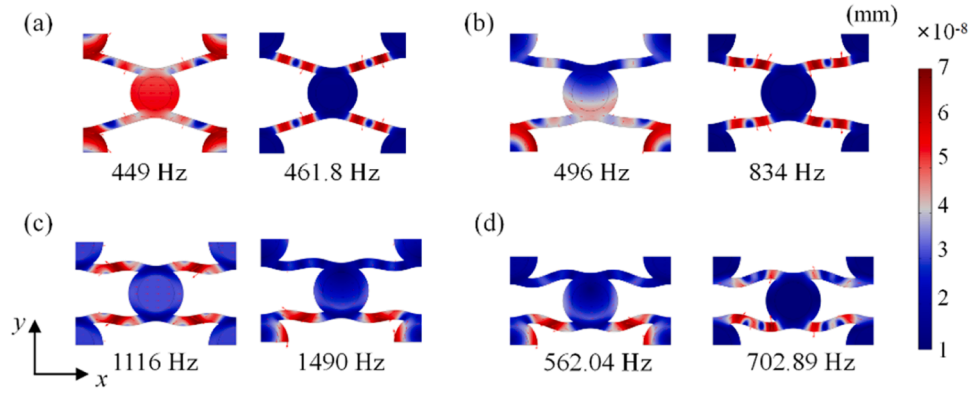


Fig. 12. Vibration modes of CCAM under different strain: (a) 0.1, (b) 0.2, (c) 0.3, and (d) 0.4.

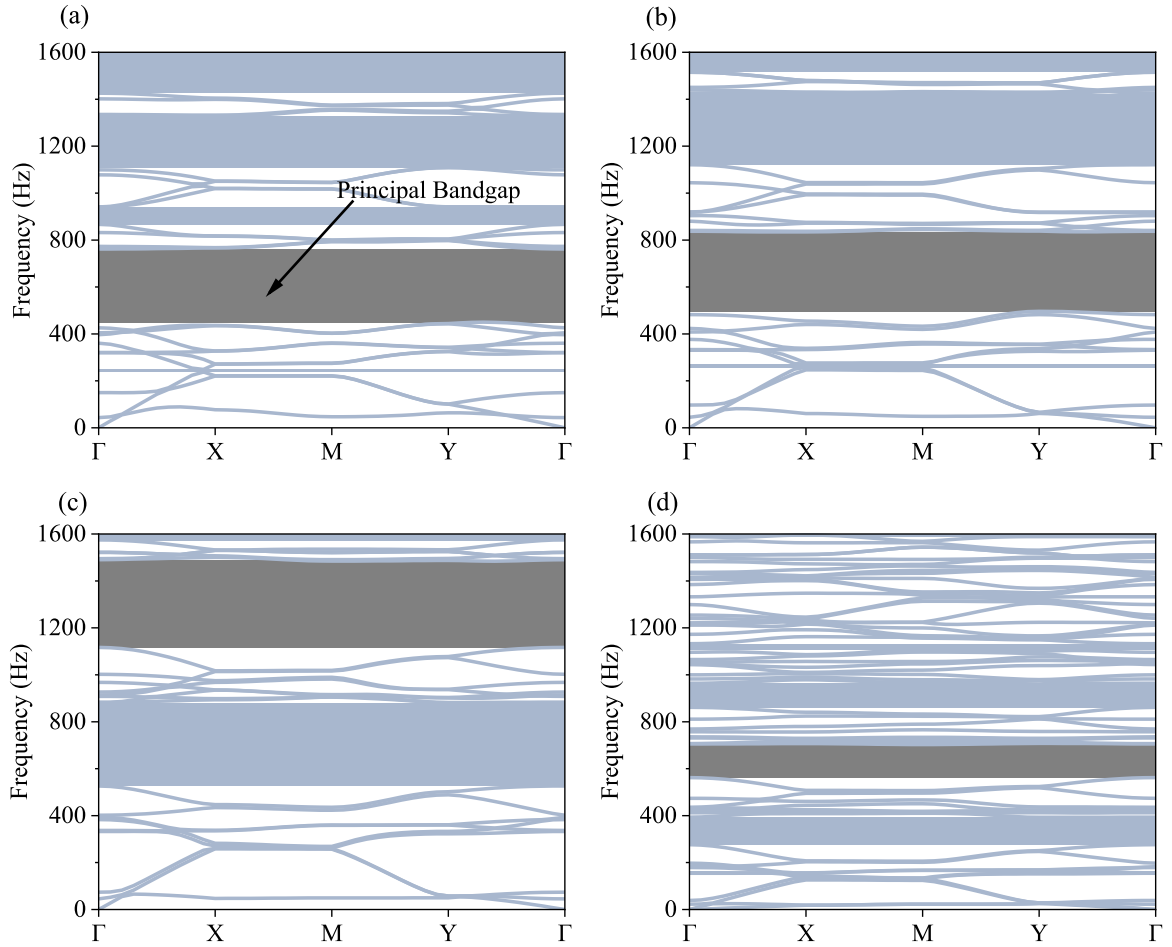


Fig. 13. Bandgap distribution of CCAM under different strain: (a) 0.1, (b) 0.2, (c) 0.3, and (d) 0.4.

generalization when handling unencountered design configurations.

Similarly, the prediction error for the core indicator $\Delta\omega_{avg}$ is maintained within an acceptable range. Although an isolated sample exhibits a maximum deviation of approximately -165.92 Hz, the absolute errors for the vast majority of predictions remain below 80 Hz. Besides, the test set yields an *RMSE* of 37.13 and a *MAPE* of 9.26%, statistically confirming strong agreement between predicted and actual values. Unlike the localized characteristics of individual bandgaps, $\Delta\omega_{avg}$ directly determines the overall bandwidth and average performance level of the vibration reduction effect within the target frequency range (0–3000 Hz). As a result, the achieved accuracy of the model may pro-

vide a reliable basis for the effective screening of wide-band insulation structures with superior performance during the preliminary design phase.

The model also demonstrates precise predictive capability for both $\Delta\omega_{nom}$ and *BGI*, achieving an R^2 value of 0.991 for $\Delta\omega_{nom}$ while maintaining *RMSE* and *MAPE* values of 48.26 and 6.63% respectively for *BGI*. This high prediction accuracy enables reliable quantification of the principal bandgap's relative width and effective evaluation of the combined performance of bandgap width and spectral concentration. The well-controlled prediction errors allow for efficient identification of optimal structures characterized by the broadest principal bandgap,

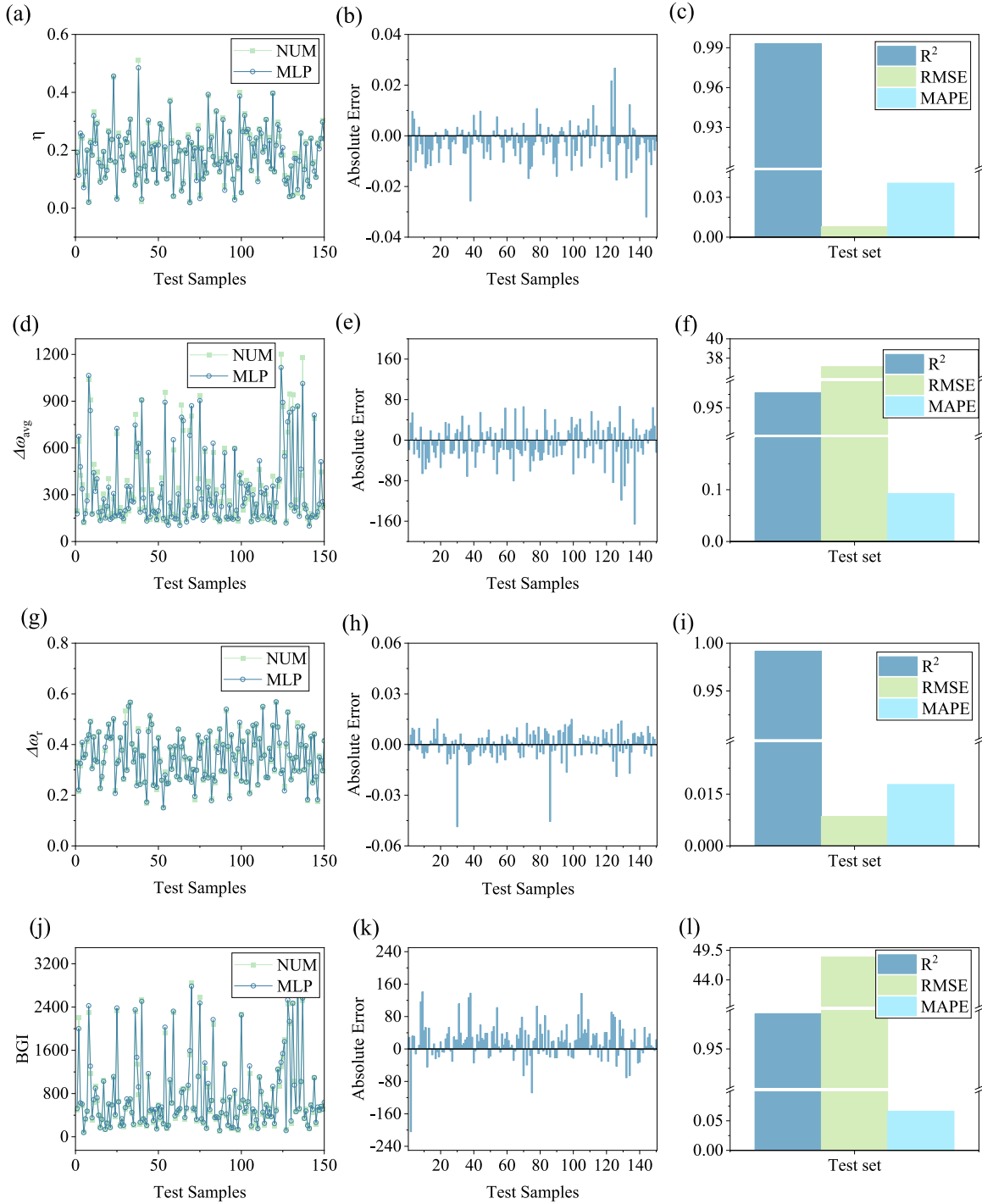


Fig. 14. Assessment of the MLP model for predicting bandgap metrics. Predicted versus numerical values for (a) η , (d) $\Delta\omega_{avg}$, (g) $\Delta\omega_{nom}$, and (j) BGI; absolute errors of predictions for (b, e, h, k); and quantitative performance metrics (e.g., R^2 , RMSE) for (c, f, i, l).

while also facilitating effective ranking and screening of candidate designs that achieve an optimal balance between wide bandgap coverage and high spectral compactness.

In summary, the model's predictive capability for all bandgap metrics has been systematically verified, demonstrating both accuracy and generalization capacity. The comparative evaluation on the test set reveals a statistically robust alignment between the model's predictive outputs and the high-fidelity simulation benchmarks, thereby validating the model's capability to deliver accurate and reliable projections of their bandgap characteristics. A data-driven foundation is thereby established for rapid design of low-frequency vibration metamaterials.

This methodology offers distinct advantages in addressing dynamic design modifications and complex performance requirements, while its implementation yields substantial gains in design efficiency through markedly reduced reliance on computationally intensive simulations during preliminary design phases.

3.6. Correlations

Fig. 15 presents the correlation analysis among key geometric parameters (t , d , h), coupling temperature T , and resulting bandgap characteristics (η , $\Delta\omega_{avg}$, $\Delta\omega_{nom}$, and BGI). As shown in Fig. 15, T exerts

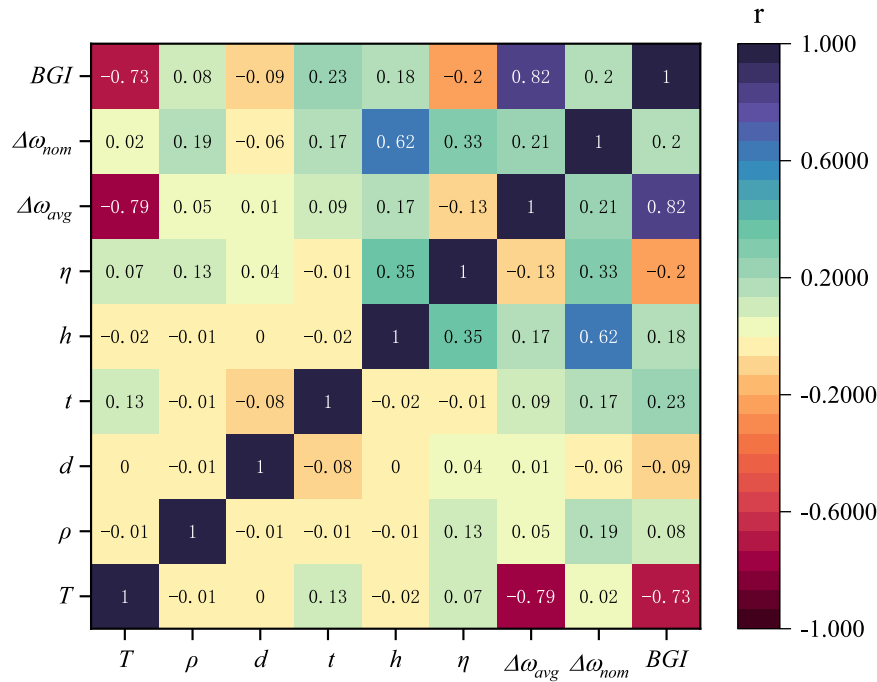


Fig. 15. Corrections heat map between design variables and bandgap metrics.

a significant influence on the vibration response, manifested in its significant negative correlations with the $\Delta\omega_{avg}$, which has a correlation coefficient of -0.79 . This observed negative correlation is primarily attributed to thermal softening, which reduces structural stiffness at elevated temperatures. This softening effect directly narrows the vibration isolation bandwidth by diminishing the width of individual bandgaps, thereby weakening the overall vibration isolation tolerance. However, the change in the number of bandgaps below 3000 Hz remains unclear. The BGI likewise demonstrates a significant negative correlation with temperature, yielding a correlation coefficient of -0.73 . This statistical relationship originates from the synergistic effect of narrowing bandgap widths and shifting central frequencies, which leads to increased dispersion in bandgap distribution and wider spacing between adjacent bandgaps. Consequently, the capability for continuous broadband vibration isolation is substantially compromised. This dispersive effect, however, can be leveraged as a design feature demonstrating the utility of thermal tuning. Such a strong, highly predictable correlation between temperature and performance indicators may thus provide a physical basis for active thermal management of bandgaps. By precisely controlling the temperature field, bandgap characteristics such as the central frequency can be directionally shifted toward a desired range.

In contrast, the h exhibits positive correlations with all four-bandgap metrics, suggesting its increase has a broad beneficial effect on vibrational performance. Most notably, its correlation with the h reaches 0.62 , indicating that h is pivotal in enhancing the relative width of the primary bandgap, beyond simply shifting its location in the frequency spectrum. This is attributed to its ability to effectively and simultaneously shift both ω_{max}^u and ω_{max}^l of the principal bandgap, thereby significantly altering its absolute bandwidth while maintaining a relatively stable central frequency. A distinct positive correlation coefficient (~ 0.35) is also observed between h and η which may imply that the total proportion of the frequency spectrum covered by bandgaps is effectively increased by a larger h . It thus suggests that an increase in h not only widens key isolation intervals such as the primary bandgap but may also promote the broadening of existing secondary bandgaps or the emergence of new ones. Consequently, the unique role of h is highlighted among geometric parameters, as it may direct bandgap positioning while also shaping their widths and distribution, thereby defining the

system's overall isolation capability.

In comparison, the vibrational performance of the CCAM is less affected by all other geometric parameters than by temperature. Selective effects are identified in t , which shows influence factors of 0.23 on BGI and 0.17 on $\Delta\omega_{nom}$, while negligible impact is seen on $\Delta\omega_{avg}$ and η . For the rib thickness t , only minor influence is registered across all four indicators, suggesting that in global optimization, it can serve as an auxiliary design variable for tailoring structural responses. In contrast, the bead diameter d holds relatively limited independent tuning value, making it suitable as an ideal tolerance compensation variable. It can be used to absorb manufacturing uncertainties or to implement risk-free performance fine-tuning in the final design stages, thereby enhancing the engineering practicality of the system.

In summary, the bandgap behavior of the CCAM is predominantly governed by temperature T , with individual bandgaps systematically narrowed and their spectral layout dispersed through thermal softening. Beyond thermal effects, the principal bandgap is simultaneously positioned and widened by h , which is identified as the decisive geometric factor. Localized adjustments can be achieved through fine-tuning t , while manufacturing compensation is suitably addressed by setting d . Collectively, hierarchical control over vibration performance is enabled through this complementary set of parameters, establishing a coherent framework for designing adaptive isolation systems.

3.7. Experimental verifications

Fig. 16 demonstrates a comparative analysis of the CCAM's vibrational performance through experimental tests and numerical simulations. All tests were conducted under an isothermal condition of 85°C , comparing the initial geometry with a strained configuration of approximately 0.4 . According to Fig. 16(a), a general consistency between experimental and simulation results is observed under both strain states, validating the effectiveness of the designed CCAM in low-frequency vibration suppression. Specifically, the primary bandgap is located within the frequency range of 414 Hz to 711.27 Hz in Fig. 16(a). Across this band, the FRF obtained from experiments and simulations exhibit highly consistent attenuation properties, with the bandgap boundaries showing close alignment. Despite localized amplitude

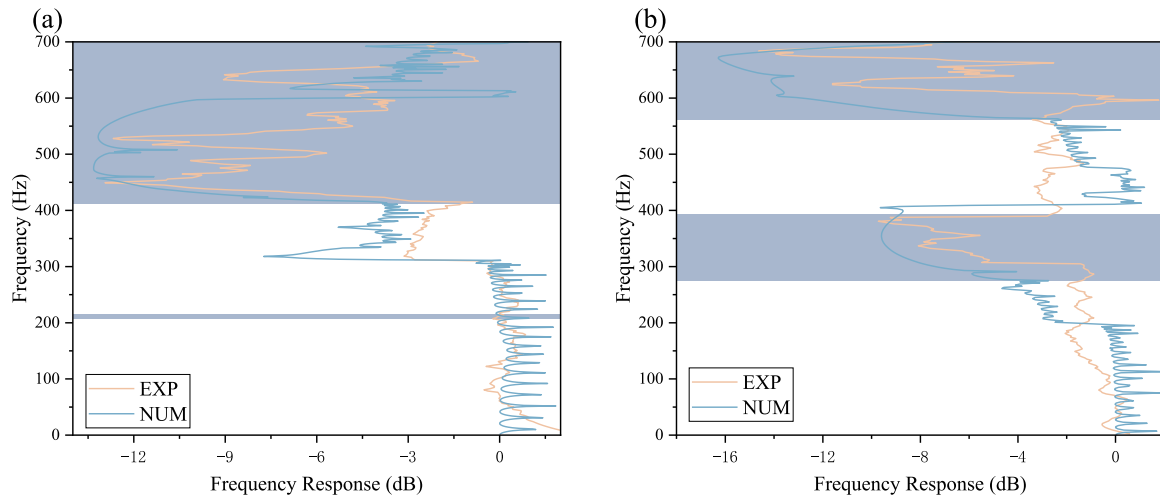


Fig. 16. Frequency response of CCAM at 85°C:(a) initial geometry, and (b) under a strain of 0.4.

variations at specific frequencies, the experimental and numerical results collectively validate the basic vibration suppression mechanism in this bandgap region. For example, the FRF values throughout the bandgap remain stably below 0 dB. A minimum FRF of -12.94 dB is recorded experimentally at 449.45 Hz, while the simulation also demonstrates pronounced attenuation of -12.28 dB at the same frequency, further corroborating the predictive reliability of the numerical model.

In contrast, within the frequency region below 300 Hz, positive FRF values are consistently observed in both experimental and simulation results, indicating limited vibration suppression capability of the CCAM in this range. A narrow bandgap observed between 206.99 Hz and 214.87 Hz is attributed to a scale mismatch between the 4×4 finite-unit experimental specimen and the periodic boundary conditions employed in the simulation. The boundary conditions inherent to an infinite periodic structure cannot be fully reproduced by a finite specimen in this work, resulting in substantial alterations being observed in the response characteristics of local resonant modes.

Fig. 16(b) presents the experimental and simulated results of the reconfigured CCAM at 0.4 strain. It may also demonstrate strong agreement in overall trends. Within the two typical bandgap regions (276.07–393.49 Hz and 562.04–702.89 Hz), the FRFs from both experimental and numerical data maintain negative values, confirming effective suppression of elastic wave propagation in these spectral ranges. This result verifies strain-induced structural reconfiguration as an effective mechanism for bandgap performance modulation.

3.8. Potential application

The proposed cementitious metamaterial can function as an adaptive low-frequency vibration suppression device. In practice, it can be embedded into primary structural members or attached to structural surfaces, serving as a pedestal, base, or isolation layer. This makes it particularly suitable for vibration-sensitive applications, such as foundations of large precision instruments, laboratory equipment, and optical testing platforms. Moreover, it is also promising for mitigating low-frequency vibration induced by road traffic, rail transit, and construction sites in urban environments as mentioned in our prior works [37].

Critically, unlike conventional thermally controlled cement-based components that require continuous heating, the proposed metamaterial operates via a triggered, on-demand adaptive mechanism. Specifically, heating is activated only when vibrations occur or exceed a prescribed threshold, enabling real-time adjustment of the effective stiffness, damping, or local configuration of the metamaterial. This, in turn, induces an adaptive shift of the bandgap, allowing the attenuation range to be tuned to the target low-frequency excitations. Such an event-

driven strategy avoids high energy consumption associated with prolonged heating in traditional solutions, resulting in a low-energy vibration mitigation system. Furthermore, the thermal actuation is largely confined to the metamaterial itself and does not significantly influence the protected primary structure, thereby ensuring effective vibration suppression while maintaining structural safety and service performance.

4. Conclusion

The 4D-printed cementitious acoustic metamaterial, namely, CCAM, is presented in this work. The specific geometric arrangement of cement and SMP components provides the structure with global shape programmability, thereby unlocking its potential for active low-frequency vibration mitigation. Additionally, a MLP-based forward predictive model was established to facilitate the high-throughput design and performance optimization of the metamaterial. The main conclusion can be summarized as follows:

The proposed CCAM achieves temperature-driven bandgap migration through local resonance. Its bandgap shifts downward with increasing temperature while the width remains constant, due to the targeted softening of the SMP-based resonant elements. This stable-width tuning capability enables robust adaptive vibration control across varying thermal environments.

Active bandgap programming in the CCAM is achieved by locking reconfigured shapes through a thermal cycle, allowing for on-demand wave manipulation. A higher applied strain of 0.4 completely reorganizes the vibrational mode into a new pattern. In contrast to the 312.8 Hz bandgap at 0.1 strain, this reconfiguration notably reduces the principal bandgap frequency to about 140.85 Hz, demonstrating significant tunability. The experimental results on transmission loss verify the strain-induced programmability of CCAM.

The established MLP forward prediction model accurately maps key design variables (t, d, h, T) to critical bandgap metrics, such as η , $\Delta\omega_{avg}$, $\Delta\omega_{nom}$, and BGI . The model demonstrates high predictive accuracy and robust generalization capability, as rigorously validated through a 5-fold cross-validation process, which yielded exceptionally low *RMSE* and *MAPE*.

Pearson correlation analysis quantitatively confirmed the distinct roles of these parameters. The analysis verified that T broadly narrows and disperses bandgaps, while h specifically positions and widens the principal bandgap, with t and d enabling localized fine-tuning. These findings confirm T as an essential input variable for accurately replicating thermal-tuning effects in the forward model.

Building on prior work [25], this study introduces temperature as a

key variable and a forward prediction model, thereby expanding the design freedom for metamaterials. However, practical challenges remain for 4D printed cementitious metamaterials, particularly regarding material compatibility [49] and long-term durability. A major hurdle is the limited variety of SMPs available, which cannot ideally adapt to the growing range of advanced cementitious materials [50,51]. Developing new, diverse SMPs is therefore an urgent need for the field.

CRediT authorship contribution statement

Yu Zheng: Writing – review & editing, Project administration, Funding acquisition. **Jianchao Zhang:** Writing – review & editing, Funding acquisition. **Ying Gao:** Writing – review & editing, Funding acquisition. **Jinrui Cao:** Formal analysis. **Jiayi Hu:** Validation, Formal analysis. **Peng Dong:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization, Funding acquisition, Validation. **Wan-qian Li:** Writing – original draft, Methodology, Investigation, Formal analysis. **Gui Li:** Writing – review & editing, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve the language and readability of the manuscript. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.conbuildmat.2026.146518](https://doi.org/10.1016/j.conbuildmat.2026.146518).

Data availability

Data will be made available on request.

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